IT VILLAGE

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## NETWORK CLAN

# OCTOBER DESIGN CHALLENGE

### BEGINNER TRACK

**PARTICIPANT WORK TEMPLATE**
**Use this workbook to document your design decisions as you work through the challenge. Complete each section in order. You may add pages, diagrams, tables or screenshots where necessary.**

| Participant / Team | Date | Submission Version |
|---|---|---|
| Olajide Olamide Basit | 07 October 2026 | v1.0 (final) |

## How to Use This Template

- Work through the sections in the same order as the challenge requirements.
- Do not jump directly into drawing the topology. First identify the business problems and requirements.
- Record the reason behind important design decisions.
- Use the tables as working documents; add rows where needed.
- You are not required to use the example IP ranges, VLAN IDs, or device choices shown anywhere in this template.
- Your final topology should agree with the addressing, segmentation, security and routing decisions documented here.

### Design Process

**1. Understand → 2. Plan → 3. Segment → 4. Address → 5. Secure → 6. Scale → 7. Validate**

## STAGE 1 — Understand the Problem

Before designing anything, translate the scenario into technical requirements.

#### 1.1 What are the main problems with the current network?

| # | Business / Technical Problem | Why It Matters | Priority |
|---|---|---|---|
| 1 | Network slows down when cameras are recording | Continuous video streams share the same flat network as staff traffic, so work applications suffer | High |
| 2 | Guests use the same Wi-Fi password as employees | Visitors sit on the trusted network and can reach staff devices and the NAS; the password also spreads | High |

| | | | |
|---|---|---|---|
| 3 | NAS/server reachable from every device | Any infected or careless device can reach company data; no least-privilege control | High |
| 4 | Cameras not isolated from employee devices | Cheap IoT cameras are common attack targets; a compromised camera must not reach staff PCs | High |
| 5 | No dedicated network management strategy | No management VLAN, addressing plan or secure admin access; devices hard to maintain and secure | Medium |
| 6 | Single ISP link: outage takes the whole office offline | Single point of failure; the business stops working when the ISP drops | High |
| 7 | Growth from 15 to ~30 employees in two years | Design must absorb more users, ports, APs and cameras without being rebuilt | Medium |

### 1.2 What must the redesigned network achieve?

| Requirement | How will your design address it? | Where will this appear in your design? |
|---|---|---|
| Isolate guests from internal resources | Separate Guest VLAN 20 + Guest SSID; firewall allows Internet only; client isolation on | Stages 4, 7, 8; VLAN 20 in topology |
| Isolate cameras and stop recording slowing the LAN | Cameras on their own VLAN 30; traffic only goes camera ↔ NAS; no user traffic mixed in | Stage 4; VLAN 30; Stage 10 |
| Protect the NAS | NAS in its own Servers VLAN 50; only Employees (SMB 445) and the camera stream pull are allowed | Stage 7 policy table; VLAN 50 |
| Manage devices securely | Management VLAN 40; SSH/HTTPS only from the admin PC; no management over Wi-Fi | Stage 7.4; VLAN 40 |
| Keep Internet up when the ISP fails | Dual-WAN on the router/firewall: WAN1 primary, WAN2 (LTE/5G or second ISP) automatic failover | Stage 6.4; WAN1/WAN2 in topology |
| Support growth to ~30 employees and 10 cameras | /24 employee subnet, /26 camera subnet, spare PoE ports, second switch and extra AP planned | Stages 5, 9, 10 |
| Keep the design simple and explainable | One router/firewall + one managed PoE switch + two APs; no unnecessary equipment | Stage 2, final summary |

# STAGE 2 — Inventory & Device Roles

Identify the devices in the scenario and decide where each belongs in your architecture.

| Device / Component | Quantity | Wired / Wireless | Proposed Network / VLAN | Role / Notes |
|---|---|---|---|---|
| ISP connection | 1 (+1 backup) | Wired hand-off | WAN1 (outside) | Primary ISP link. A second WAN (LTE/5G or other ISP) is added as backup. ISP router set to bridge/modem mode to avoid double NAT. |
| Router / Firewall | 1 | Wired | Gateway for all VLANs (trunk to SW1) | Routing, NAT/PAT, DHCP, DNS forwarder, inter-VLAN routing, firewall. Mgmt IP 10.10.40.1 |
| Managed switch | 1 now (2nd at 10 cameras) | Wired | Trunk to router; access ports per VLAN | 24-port PoE+ with 802.1Q VLANs, DHCP snooping, port security. Mgmt IP 10.10.40.2 |
| Wireless APs | 2 | Wired uplink (PoE), wireless service | Mgmt VLAN 40 native; SSIDs on VLAN 10 and 20 | One per floor. Static mgmt IPs 10.10.40.11 and .12 |
| Employee PCs | 8 | Wired | VLAN 10 Employees | Access ports; DHCP from router |
| Employee laptops | 7 | Wireless | VLAN 10 via TechPoint-Staff SSID | DHCP from router |

| | | | | |
|---|---|---|---|---|
| Printers | 2 | Wired | VLAN 60 Printers | Static 10.10.60.11 and .12; only Employees may print |
| IP cameras | 4 | Wired (PoE) | VLAN 30 Cameras | DHCP reservations .11–.14; no Internet; NAS pulls the streams |
| NAS / Server | 1 | Wired | VLAN 50 Servers | Static 10.10.50.10; recording storage + file shares |
| Other | 1 admin PC + UPS | Wired | VLAN 40 Management | Admin PC 10.10.40.20 for device administration; UPS for router, switch and NAS |

**2.1 Which device will act as the default gateway for the LAN networks?**

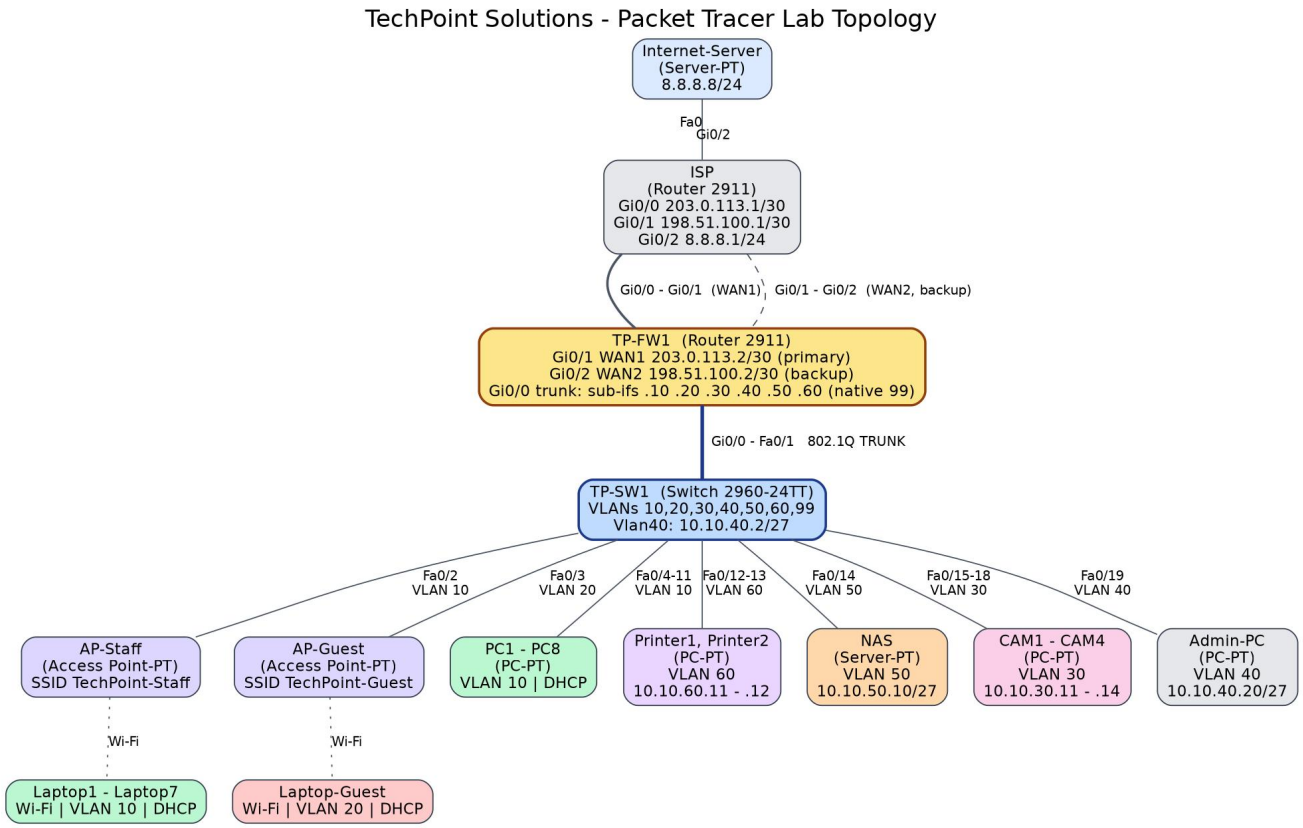
The router/firewall (TP-FW1) is the default gateway for every VLAN. It has one sub-interface per VLAN on the 802.1Q trunk to SW1 (router-on-a-stick): 10.10.10.1, 10.10.20.1, 10.10.30.1, 10.10.40.1, 10.10.50.1 and 10.10.60.1. The switch only does Layer 2, so all hosts point to their .1 address.

**2.2 Which device(s) will perform routing, NAT, DHCP and firewall functions?**

TP-FW1 performs all four functions: routing (between VLANs and towards the ISP), NAT/PAT on the WAN interface(s), DHCP for VLANs 10, 20 and 30 (static addressing for 40/50/60), and the stateful firewall rules between VLANs. One device keeps the design simple for a 15-person office and is easy to justify. The ISP router is put in bridge/modem mode so NAT happens only once (no double NAT).

# STAGE 3 — Design the Network Topology

Create your logical topology below or insert your completed diagram on a separate page.




**3.1 Explain the flow from the Internet to an employee device.**

Internet → ISP1 (WAN2 as backup) → TP-FW1 outside interface (stateful firewall + NAT/PAT) → TP-FW1 inside trunk → TP-SW1 → access port in VLAN 10 (wired PC) or AP1/AP2 (Wi-Fi laptop) → employee device. Return traffic follows the same path. A PC sends to its gateway 10.10.10.1, the firewall checks the rules, NATs the packet to the WAN address and forwards it to the ISP.

### 3.2 How do the wireless access points connect to the network?

Each AP is powered by PoE from SW1 and connects to a trunk port: native VLAN 40 for the AP's own management IP (10.10.40.11 and .12) and VLANs 10 and 20 tagged. This lets TechPoint-Staff map to VLAN 10 and TechPoint-Guest to VLAN 20 over one cable. One AP per floor gives coverage on both floors.

### 3.3 Where does the NAS connect, and why?

The NAS connects to an access port on SW1 in VLAN 50 (Servers) with static IP 10.10.50.10. Because it is in its own VLAN behind the firewall, every connection to it must pass the firewall rules: Employees can reach it, the NAS can pull camera streams, and guests are blocked. It also stops the NAS being reachable from every device.

## STAGE 4 — Network Segmentation

Break the flat network into logical networks. You should have a clear reason for every segment.

| VLAN / Network ID | Name | Purpose | Who / What Uses It? | Should It Access Other Networks? | Notes |
|---|---|---|---|---|---|
| 10 | EMPLOYEES | Staff workstations and laptops | 8 PCs + 7 laptops (up to ~30 staff) | Internet: yes. NAS: SMB. Printers: yes. Cameras: only a named viewer PC. Mgmt: no | Main user VLAN; Staff SSID maps here |
| 20 | GUEST | Visitor Internet access | Visitor phones/laptops | Internet only. Nothing internal | Client isolation, bandwidth cap, separate password |
| 30 | CAMERAS | IP camera / IoT traffic | 4 cameras (10 after twist) | None initiated by cameras. NAS may pull streams | No Internet; keeps video traffic away from users |
| 40 | MGMT | Administration of network gear | Router, switches, APs, admin PC | Admin PC may reach device management interfaces | Static IPs, no DHCP, no SSID |
| 50 | SERVERS | Protected data storage | NAS (and future servers) | Accepts Employees on SMB; NAS may pull from Cameras | Additional VLAN: controlled access to data |
| 60 | PRINTERS | Network printers | 2 printers | Accept print jobs from Employees only | Additional VLAN: printers are weak devices |

### 4.1 Why should employees, guests, cameras/IoT and management be separated?

Each group has a different trust level and traffic type. Employees are trusted. Guests are untrusted and only need the Internet. Cameras are low-security IoT devices that produce constant video traffic, so isolating them protects staff and stops recording traffic competing with work traffic. Management carries admin access to the router, switch and APs and must be reachable only by the admin. Separate VLANs give separate broadcast domains and one place (the firewall) to control access.

### 4.2 Are you creating any additional VLANs? If yes, why?

Yes, two. VLAN 50 Servers puts the NAS behind firewall rules instead of being open to every device. VLAN 60 Printers stops printers (often old firmware, many open ports) being a stepping stone and lets only Employees print. Unused switch ports are also shut down and parked in VLAN 99 (no IP) as a security measure.

# STAGE 5 — IP Addressing Plan

Complete an IPv4 addressing plan that matches your VLAN design. Make sure every subnet is unique.

| Network / VLAN | Purpose | Network Address | Prefix / Mask | Usable Host Range | Default Gateway | DHCP Range | Reserved / Static Addresses |
|---|---|---|---|---|---|---|---|
| VLAN 10 Employees | Staff devices | 10.10.10.0 | /24 255.255.255.0 | 10.10.10.1 – 10.10.10.254 | 10.10.10.1 | 10.10.10.50 – .250 | Gateway .1; .2–.49 kept for static/reserved needs |
| VLAN 20 Guest | Visitor Wi-Fi | 10.10.20.0 | /24 255.255.255.0 | 10.10.20.1 – 10.10.20.254 | 10.10.20.1 | 10.10.20.10 – .250 | Gateway .1 only (short lease, 2 hrs) |
| VLAN 30 Cameras | IP cameras / IoT | 10.10.30.0 | /26 255.255.255.192 | 10.10.30.1 – 10.10.30.62 | 10.10.30.1 | 10.10.30.31 – .62 | Camera reservations .11–.30 (10 now, up to 20) |
| VLAN 40 Mgmt | Device management | 10.10.40.0 | /27 255.255.255.224 | 10.10.40.1 – 10.10.40.30 | 10.10.40.1 | None (static) | Router .1, SW1 .2, SW2 .3, APs .11–.19, admin PC .20 |
| VLAN 50 Servers | NAS / servers | 10.10.50.0 | /27 255.255.255.224 | 10.10.50.1 – 10.10.50.30 | 10.10.50.1 | None (static) | NAS .10; spare .11–.19 for future server |
| VLAN 60 Printers | Network printers | 10.10.60.0 | /27 255.255.255.224 | 10.10.60.1 – 10.10.60.30 | 10.10.60.1 | None (static) | Printers .11 and .12; spare .13–.19 |

### 5.1 Why did you choose this subnet size?

A /24 for Employees and Guest (254 hosts) is easy to read and gives plenty of room. A /26 for Cameras (62 hosts) is far more than 10 cameras but leaves room to double. A /27 (30 hosts) for Management, Servers and Printers is enough for small fixed device counts. All subnets come from 10.10.0.0/16 with the VLAN ID as the third octet, so VLAN and subnet are easy to match, and nothing overlaps.

### 5.2 How does your addressing plan support growth from 15 to approximately 30 employees?

The employee /24 has about 250 usable addresses. 30 employees with around three devices each (PC, laptop, phone) is roughly 90 addresses, so there is more than double the headroom and no renumbering is needed. The DHCP pool (.50–.250) already covers it. Only switch ports and AP capacity need to grow.

### 5.3 Which devices should use static IPs or DHCP reservations? Why?

Static or reservation: the router, switches and APs (static in VLAN 40), the NAS (10.10.50.10 static), the printers (.11 and .12 static, because users' print queues point to fixed addresses), cameras (DHCP reservation by MAC, .11–.30, so the NAS and firewall rules always find them) and the admin PC. Staff PCs, laptops, phones and guest devices use normal dynamic DHCP.

# STAGE 6 — DHCP, Routing & Internet

### 6.1 Where will DHCP run?

DHCP runs centrally on the router/firewall, with one pool per VLAN (10, 20, 30). The switch runs DHCP snooping with only the router-facing trunk trusted, so a rogue DHCP server on an access port is dropped. VLANs 40, 50 and 60 use static addresses.

### 6.2 What is the default gateway for each VLAN?

| VLAN / Network | Default Gateway | DHCP Provided By | DNS Provided By |
|---|---|---|---|
| VLAN 10 Employees | 10.10.10.1 (router sub-interface) | Router/firewall | Router DNS forwarder → 1.1.1.1 / 8.8.8.8 |
| VLAN 20 Guest | 10.10.20.1 | Router/firewall | Public DNS handed out (1.1.1.1 / 9.9.9.9); no internal DNS |
| VLAN 30 Cameras | 10.10.30.1 | Router/firewall (MAC reservations) | Router (not needed; no Internet) |
| VLAN 40 Mgmt | 10.10.40.1 | None – static | Router DNS forwarder |
| VLAN 50 Servers | 10.10.50.1 | None – static | Router DNS forwarder |
| VLAN 60 Printers | 10.10.60.1 | None – static | Router DNS forwarder |

### 6.3 Where will NAT/PAT occur?

NAT/PAT occurs on the router/firewall's outbound WAN interface(s) (PAT/overload) for the VLANs allowed to go out (Employees and Guest). It happens once only because the ISP device is in bridge mode. Cameras are not NATed because they are denied Internet access.

### 6.4 What device holds the default route toward the ISP?

TP-FW1 holds the default route 0.0.0.0/0. The primary route points to ISP1 (WAN1); a floating backup default route (higher administrative distance/metric) points to ISP2 (WAN2). If a health check to ISP1 (ping to the ISP gateway/1.1.1.1) fails, traffic switches to WAN2 automatically.

### 6.5 How will traffic move between VLANs?

Inter-VLAN routing is done by TP-FW1 using router-on-a-stick: one 802.1Q trunk to SW1 with a sub-interface per VLAN. Traffic between VLANs goes up to the router, is checked against the firewall rules and is routed back down the trunk. That makes the firewall the single control point. If the trunk becomes a bottleneck, a Layer 3 switch can take over inter-VLAN routing later.

## STAGE 7 — Security & Access Control

Define what each network is allowed and not allowed to reach. Think in terms of least privilege.

| Source | Destination | Allow / Deny | Protocol / Service (if known) | Reason |
|---|---|---|---|---|
| Guest | Internet | Allow | TCP 80/443, TCP/UDP 53 | Visitors only need web access |
| Guest | Employee network | Deny | Any | Protect staff devices |
| Guest | NAS | Deny | Any | Protect company data |
| Guest | Cameras / IoT | Deny | Any | Protect cameras and footage |
| Guest | Management | Deny | Any | Protect network devices |
| Employees | NAS | Allow | SMB TCP 445 only | Staff need file shares; only the port required |
| Employees | Cameras / IoT | Deny (except named viewer PC) | Viewer PC: TCP 554 / 443 | Least privilege; footage viewed through NAS or one viewer PC |
| Cameras / IoT | Employees | Deny | Any (new connections) | Cameras never initiate; replies to allowed sessions only |
| Management | Network devices | Allow | SSH 22, HTTPS 443, SNMPv3 161 | Secure administration from the admin PC only |
| NAS | Cameras / IoT | Allow | RTSP TCP 554, ONVIF/HTTP(S) 80/443 | NAS pulls the streams, so cameras need no outbound access |
| Employees | Printers | Allow | TCP 9100, TCP 631 | Staff can print |
| Cameras / IoT | Internet | Deny | Any | No cloud call-home or data leaving the site |
| Any other network | Router / switch / AP management | Deny | Any | Management only from VLAN 40 |
| Default (any) | Any | Deny + log | Any | Default-deny: anything not listed is blocked |

### 7.1 What should happen if a guest tries to reach an employee PC?

It is blocked. The firewall drops traffic from VLAN 20 to VLAN 10 (explicit deny rule, default-deny behind it) and logs it. Client isolation on the guest SSID also stops guests seeing each other.

### 7.2 What should happen if a guest tries to reach the NAS?

It is blocked. VLAN 20 to VLAN 50 is denied and logged. The NAS only accepts SMB from the Employees VLAN and its own camera connections.

### 7.3 What should happen if a camera tries to initiate a connection to an employee PC?

It is denied. Cameras never initiate connections: the stateful firewall drops any new connection from VLAN 30 to Employees (and the Internet). The NAS pulls the streams, so only replies are allowed. A camera trying to connect out would be logged and treated as a sign of compromise.

### 7.4 How will network devices and APs be securely managed?

Devices are managed only from VLAN 40, using SSH v2 and HTTPS (Telnet and plain HTTP disabled), from the admin PC only. Controls: default passwords changed, unique strong admin passwords with named accounts and lockout, SNMPv3, NTP and syslog, regular firmware updates and config backups, unused ports shut down in VLAN 99, DHCP snooping and port security. No management over Wi-Fi.

# STAGE 8 — Wireless Design

| SSID | Purpose | VLAN / Network | Who Can Connect? | Internet Access | Internal Access | Security / Notes |
|---|---|---|---|---|---|---|
| TechPoint-Staff | Employee Wi-Fi | VLAN 10 | Employees only | Yes | NAS, printers per policy | WPA2/WPA3-Personal (AES), long unique passphrase (14+ chars). Upgrade path: WPA2-Enterprise (802.1X/RADIUS) |
| TechPoint-Guest | Visitor Wi-Fi | VLAN 20 | Visitors | Yes (bandwidth-capped) | None | Different passphrase (rotated, or voucher/captive portal); client isolation ON |
| TechPoint-IoT (reserved, off) | Future wireless IoT devices | VLAN 30 | Approved devices only | No | None | Not broadcast now; only created if wireless IoT is ever needed |

**8.1 How will guest wireless be isolated from internal resources?**

The guest SSID maps to VLAN 20, which has its own subnet. The firewall allows VLAN 20 to the Internet only (80/443/53) and denies all private ranges and the other VLANs. AP client isolation is on, the passphrase differs from staff Wi-Fi, bandwidth is capped and guests use public DNS.

**8.2 How will the employee wireless network map to the correct VLAN?**

On each AP the TechPoint-Staff SSID is tagged to VLAN 10. The AP uplink is a trunk on SW1 carrying VLANs 10 and 20 tagged and 40 native. A staff laptop should receive a 10.10.10.x address from the router's VLAN 10 DHCP pool; I would confirm this with ipconfig.

# STAGE 9 — Scalability

Now test your design against the expected growth.

| Growth Item | Current | Future Requirement | Will Your Design Handle It? | What Changes Would Be Needed? |
|---|---|---|---|---|
| Employees | 15 | ~30 | Yes | None for IP space: /24 has ~250 addresses (~90 devices expected). Add ports and AP capacity. |
| IP cameras | 4 | 10 | Yes | /26 holds 62 hosts. Add second switch and check PoE budget. |
| Wireless clients | ~20 (7 laptops + phones) | ~45–60 Wi-Fi devices | Mostly | Add a third AP, use 5 GHz, plan channels, cap guest bandwidth. |
| Switch ports | 18 of 24 used | 35+ endpoints | No (one switch) | Add second 24-port PoE switch (SW2), trunk/LAG to SW1; later a 48-port core. |
| Access points | 2 | Potentially more | Yes | Mgmt /27 has room for ~10 APs; add a PoE port and same SSIDs. |
| Internet traffic | Single ISP link | Higher | Partly | Check WAN bandwidth and router throughput with NAT+firewall; cap guest; dual-WAN. |
| NAS / storage | 1 NAS, 4 cameras | More camera recordings | Needs sizing | 10 cameras at ~4 Mbps ≈ 430 GB/day continuous. Size disks (RAID), set retention, use motion recording / H.265. |

**9.1 What is the biggest scalability concern in your design?**

The biggest concern is switch ports and PoE: one 24-port switch is full at 10 cameras, and all inter-VLAN traffic crosses a single 1 Gbps trunk. A second concern is single points of failure (one router, one switch). A second switch, a link aggregation (LAG) uplink and later a Layer 3 switch address these.

### 9.2 What equipment or capacity would you check before deployment?

I would check the PoE power budget (watts) against APs plus cameras; switch port count and uplink speed; router throughput with NAT and firewall enabled; ISP bandwidth (especially upload); NAS disk capacity, retention time and network ports; Wi-Fi coverage with a site survey on both floors; and UPS capacity.

## STAGE 10 — The Twist: 6 Additional Cameras

The company purchases 6 additional IP cameras, bringing the total to 10.

### 10.1 What changes in your network design?

The VLAN design does not change. The 6 new cameras join VLAN 30 with reservations .15–.20 (cameras use .11–.20). I add a second 24-port PoE switch (SW2) on Floor 2, linked to SW1 by a trunk (or 2-port LAG), re-check the PoE budget, and make sure the NAS has storage and recording settings for 10 cameras. The firewall rules are unchanged because the cameras are in the same VLAN.

### 10.2 Does your camera subnet have enough IP addresses?

Yes. VLAN 30 is a /26 with 62 usable addresses. Ten cameras use .11–.20, which leaves 52 free addresses, so it has room for 20 or more cameras.

### 10.3 Do you have enough switch ports / PoE capacity?

Ports: before the twist 18 of 24 are used; with 6 more cameras all 24 are used, which leaves none free for growth, so SW2 is needed. PoE: roughly 10 cameras × 8 W (~80 W) plus 2 APs × 15 W (~30 W) = about 110–130 W. I would confirm that the switch PoE budget is above this (ideally 185 W or more) and spread cameras across SW1 and SW2.

### 10.4 Will the additional camera traffic affect the switch, uplink, firewall or NAS?

Yes, somewhat. Assuming about 4 Mbps per camera, 10 cameras produce roughly 40 Mbps (up to ~80 Mbps at higher quality) continuously. Because they are routed to the NAS (router-on-a-stick), that traffic crosses the trunk and the firewall in both directions, which a 1 Gbps trunk handles but adds load. The NAS needs enough disk write speed and about 430 GB/day of storage if recording continuously (less with motion recording). Staff traffic is protected because cameras are in their own VLAN; a Layer 3 switch can later keep this traffic off the router.

### 10.5 Would you change your VLAN design? Why or why not?

No change. The functions and policy are the same and the /26 has plenty of room. I would only change it if the camera count grew beyond about 50, or if a different type of device, such as door access control, needed separating. Cameras on both floors stay in VLAN 30 trunked across SW1 and SW2.

# STAGE 11 — Bonus Troubleshooting

Scenario: An employee laptop can access the Internet but cannot access the NAS. Another employee on the same network can access the NAS normally.

### 11.1 What is your troubleshooting sequence?

| Step | What Will You Check? | Expected Result / What Would It Tell You? |
|---|---|---|
| 1 | Define scope: compare with the working colleague (same SSID/port/VLAN?). What fails – ping by IP, \\NAS by name, or a specific app? | Shows whether it is the laptop, the network path or the NAS, and IP vs name problem |
| 2 | On the laptop: ipconfig /all (IP, mask, gateway, DNS) and the connected SSID | Expected 10.10.10.x /24, gateway 10.10.10.1. 10.10.20.x = on Guest Wi-Fi; 169.254.x.x = DHCP failure; wrong static = misconfig |
| 3 | Ping gateway 10.10.10.1, then 1.1.1.1, then the NAS IP 10.10.50.10 | Gateway OK but NAS fails = problem beyond the gateway (firewall/NAS). Gateway fails = Layer 1/2 (AP/switch/VLAN) |
| 4 | Test name vs IP and the service port: nslookup nas; Test-NetConnection 10.10.50.10 -Port 445 | Works by IP but not name = DNS/cache. Port 445 blocked = firewall rule or NAS-side block |
| 5 | Access point / switch: which SSID/port; show vlan brief; show mac address-table; trunk allowed VLANs; port-security status | Laptop MAC should appear in VLAN 10. Wrong VLAN or err-disabled port = mapping/port issue |
| 6 | Gateway: router DHCP lease for the laptop MAC; traceroute to NAS; check VLAN 10 sub-interface and ARP | Lease in wrong VLAN or traceroute stopping at a hop shows where the path breaks |
| 7 | Firewall/ACL: logs and rule hit counters for laptop IP → 10.10.50.10; check the Employees address group | Deny entries mean a rule excludes this IP (outside range, IP conflict, auto-ban) |
| 8 | NAS side: NAS logs, blocked-IP list, share permissions, user account; retry with another account | NAS auto-blocked the laptop IP or user lacks permission; clear block or fix permissions, then retest |

### 11.2 What are your top three possible causes?

| Possible Cause | Evidence That Would Support It | Test / Verification |
|---|---|---|
| Laptop on the wrong network (Guest SSID, wrong VLAN, bad static IP or IP conflict) | ipconfig shows 10.10.20.x, 169.254.x.x or a wrong gateway; Internet works but internal does not | Check SSID, ipconfig, DHCP lease table and switch MAC table; reconnect to TechPoint-Staff and retest |
| Firewall/ACL rule or NAS auto-block affecting only this laptop's IP | Gateway pings, but TCP 445 fails only from this laptop; deny entries in logs; IP in NAS block list | Test-NetConnection on 445, firewall log/hit counter, NAS blocked-IP list; correct rule or unblock |
| Name resolution / local client problem (DNS cache, hosts file, local firewall/VPN, saved credentials) | NAS reachable by IP but not by name; works for the other user; credential prompt errors | nslookup, ipconfig /flushdns, check hosts file, temporarily disable local firewall/VPN, re-enter credentials |

# STAGE 12 — Final Design Review

Use this checklist before submitting. Your final diagram and tables should tell the same story.

| Check | Complete ✓ | Notes / Fix Required |
|---|---|---|
| ISP and Internet edge are clearly shown. | ☑ | Stage 3 topology: WAN1/WAN2 to TP-FW1 |
| Router/firewall role is clearly defined. | ☑ | Stage 2.2 and 6 |
| Employee network is separated from guest traffic. | ☑ | Guest VLAN 20, Stage 7 |
| Camera/IoT network is separated from users. | ☑ | Cameras VLAN 30 |
| Management network is separated. | ☑ | Mgmt VLAN 40 |
| NAS access is controlled. | ☑ | NAS in VLAN 50, SMB from Employees only |
| IP addressing is complete and non-overlapping. | ☑ | Stage 5, no overlap |
| Every VLAN has a clear gateway. | ☑ | Router sub-interfaces .1 on every VLAN |
| DHCP design is defined. | ☑ | Router DHCP for VLANs 10/20/30 |
| NAT is defined. | ☑ | PAT on WAN1/WAN2, Stage 6.3 |
| Inter-VLAN routing is defined. | ☑ | Router-on-a-stick, Stage 6.5 |
| Guest access to internal resources is restricted. | ☑ | Guest → Internet only |
| The design supports approximately 30 employees. | ☑ | /24 gives ~250 addresses |

| | | |
|---|---|---|
| The design can accommodate 10 cameras. | ☑ | /26 gives 62 hosts, 2nd switch added |
| Wireless SSIDs are mapped to the intended networks. | ☑ | Stage 8 SSID table |
| The topology matches the addressing and security tables. | ☑ | Same VLANs/IPs in every table |
| You can explain why you made each major design decision. | ☑ | Decisions justified in each stage |

# Final Design Summary

**In 5–8 sentences, explain your overall network design and the most important decisions you made.**

I replaced the flat ISP-router network with a segmented design built around one router/firewall (TP-FW1) and one managed PoE switch (TP-SW1) with two PoE access points. Six VLANs separate Employees (10), Guests (20), Cameras (30), Management (40), Servers/NAS (50) and Printers (60), all addressed from 10.10.0.0/16 with the VLAN ID as the third octet. The router/firewall is the default gateway for every VLAN, runs DHCP, performs NAT once and enforces least-privilege rules: guests reach only the Internet, cameras never initiate connections, and only employees reach the NAS on SMB. The Staff and Guest SSIDs map to VLANs 10 and 20 over a trunk to each AP. A second WAN link (LTE/5G or second ISP) with automatic failover removes the single point of failure from the ISP. The plan reaches 30 employees without renumbering because the employee /24 has room; ports run out first, so a second 24-port switch is added when the cameras reach 10. I kept the design simple (router-on-a-stick, no Layer 3 switch) and noted where it would be upgraded as the business grows.

# Submission Checklist

- ☑ Completed topology diagram
- ☑ Completed VLAN / network table
- ☑ Completed IP addressing plan
- ☑ Completed security / access-control table
- ☑ Completed wireless design
- ☑ Completed scalability analysis
- ☑ Completed camera-growth twist
- ☑ Completed troubleshooting approach
- ☑ Final design summary

**START SMALL. DESIGN SMART.**